ATLAS Master Flightline Clock — Project Handover Dossier

UPDATED VERSION — July 26, 2026 (Session 2)

> **Document Purpose:** Comprehensive handover for any future AI assistant working on Project ATLAS.
> Read every section before touching anything. Read the user's actual files before changing anything.
> **Previous handover:** `ATLAS_PROJECT_HANDOVER_Sun_26_July.md` (earlier today — now superseded by this document)
> **System Version:** v1.6.0 — fully working on Mac and ready for Windows deployment

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CRITICAL CHANGES SINCE LAST HANDOVER

These are the most important things that changed during Session 2. Read these first.

1. Project Location Changed (IMPORTANT)

The project was moved from `~/Documents/project_atlas/` to `~/project_atlas/`.
Do not reference the old Documents path anywhere. The new canonical location is:

```
/Users/petehep/project_atlas/
```

2. iCloud Was Destroying the PySide6 Install — Now Fixed

The recurring "cocoa platform plugin not found" crash was caused by iCloud Drive syncing the `.venv` folder and creating conflicted copies of Qt `.dylib` files. Moving the project out of `~/Documents/` permanently fixed this. Full details in Section 7.

3. Windows Deployment Files Created

A `windows/` subfolder now exists inside the project containing Windows-specific launcher scripts. The Mac source files were NOT modified for this — the launchers are completely separate. Full details in Section 9.

4. Two Source Files Patched

`src/core/database.py` and `src/gui/windows.py` have been updated with cross-platform fixes. These patches are applied on the Mac. Full details in Section 5.

5. Python Version Correction

The previous handover incorrectly stated Python 3.14. The actual version is **Python 3.12**. This has been corrected throughout this document.

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1. Project Identity & Purpose

What Is ATLAS?

ATLAS is a **specialised 6-row digital display system** designed to manage and synchronise race timing, pilot insertion windows, and heat data during competitive aviation events — specifically **F3K model glider competitions**.

It syncs with the local PC clock for real-time accuracy and guides pilots through two distinct phases:
1. **Pre-Race Phase** — Countdown to track opening, displaying upcoming heat info
2. **Active-Race Phase** — Live timing showing entry windows and heat countdowns

User Context

- **User:** Peter ("petehep"), based in **Adelaide, South Australia**
- **Competition context:** F3K model glider events — multiple heats per day, each with defined track-open/track-close windows and thermalling directions
- **Target deployment:** Standalone "appliance" at a flying field — visible to pilots from a distance
- **Hardware ambitions:** Physical LED matrix panels (P5/P10) — currently using software simulator
- **Development machine:** MacBook Air (username `petehep`, macOS 15.5 Sequoia)
- **Target deployment hardware:** Windows PC (confirmed — Windows setup now complete)

Communication Style — IMPORTANT

- Peter is direct, casual, and technical. Swearing is normal, don't be precious about it.
- He values things that work over elegant theory.
- **Do not change code without showing him first.**
- **Ask before assuming** — if files seem wrong or missing, ask him.
- If you break something, own it immediately and fix it cleanly without excuses.
- Humour is fine and welcome.

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2. Current Project Location & Structure

CANONICAL LOCATION (Updated Session 2)

```
/Users/petehep/project_atlas/
```

NOT `~/Documents/project_atlas/` — that location no longer exists. The project was moved to the home folder to prevent iCloud interference.

Full Directory Structure

```
~/project_atlas/
├── .git/                          ← Git repository (active, many commits)
├── .gitignore                     ← Includes .venv/ — correct, do not remove
├── .venv/                         ← Python virtual environment (rebuilt fresh at new
location)
├── archive/                       ← Old/archived files
├── assets/                        ← Static assets
├── atlas_config.json              ← Legacy JSON config (not used by main.py, harm-
less)
├── atlas.sh                       ← Legacy startup script (may be outdated)
├── config/                        ← Config directory
├── data/
│   └── atlas.db                   ← SQLite database (heats + settings)
├── docs/                          ← Project documentation (38 items)
├── requirements.txt               ← Python dependencies
├── run_atlas.sh                   ← PRIMARY MAC LAUNCHER (updated Session 2)
├── src/
│   ├── core/
│   │   ├── __init__.py
│   │   ├── config.py              ← Legacy JSON config loader (not used, harmless)
│   │   ├── database.py            ← PATCHED Session 2 — cross-platform path fix
│   │   ├── engine.py              ← 1Hz state machine, heat phase logic (untouched)
│   │   ├── models.py              ← Pydantic data models (untouched)
│   │   └── timers.py              ← Legacy standalone timer (not used, harmless)
│   ├── gui/
│   │   ├── __init__.py
│   │   ├── fonts.py               ← 5×7 bitmap font data (untouched, correct)
│   │   ├── lcd_display.py         ← Full-resolution LCD backend (added Session 1)
│   │   ├── led_simulator.py       ← LED pixel simulator (untouched)
│   │   ├── themes.py              ← Qt CSS styling constants (untouched)
│   │   └── windows.py             ← PATCHED Session 2 — EXIT TO DESKTOP fix
│   └── main.py                    ← Application entry point (updated Session 1)
├── tests/
├── tools/
└── windows/                       ← NEW Session 2 — Windows-specific files only
    ├── .gitkeep
    ├── run_atlas.bat              ← Windows CMD launcher
    ├── run_atlas.ps1              ← Windows PowerShell launcher
    └── SETUP_WINDOWS.md           ← Windows setup guide
```

How to Launch on Mac

```
cd ~/project_atlas
./run_atlas.sh
```

The `run_atlas.sh` at the project root is the correct launcher. It handles venv activation, PYTHONPATH and Qt plugin path automatically.

If you get a permission denied error:

```
chmod +x run_atlas.sh
./run_atlas.sh
```

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3. Technical Architecture

Pattern: Model-View-Controller (MVC)

PySide6 (Qt for Python) + SQLite.

```
                  main.py
            (Entry point & wiring)
                 /          \
          CORE                GUI
    engine.py           windows.py        (Operator Console)
    database.py         led_simulator.py  (LED display)
    models.py           lcd_display.py    (LCD display)
    timers.py*          fonts.py          (bitmap font data)
    config.py*          themes.py         (CSS styling)

    * = legacy, not actively used
```

Signal/Slot Architecture

- **`engine.model_updated`** → `display.update_model(model)` — 1Hz tick
- **`engine.brightness_changed`** → `display.set_brightness(value)` — on operator slider change

AtlasDisplayModel (Pydantic — the data contract)

```python
class AtlasDisplayModel(BaseModel):
    local_time: str = "00:00:00"
    primary_timer: str = "--:--"
    primary_timer_label: str = "SYSTEM IDLE"
    primary_timer_color: str = "#888888"
    band_color: str = "#333333"
    heat_number: str = "--"
    thermalling_dir: str = "---"
    is_armed: bool = False
    planned_open: str = "--:--"
    planned_close: str = "--:--"
```

Display Backends

Two backends, selectable from Settings tab (requires restart):

Backend	ID	File	Description
LED Simulator	`"simulator"`	`led_simulator.py`	Pixel-art LED emulation, 5×7 bitmap font
LCD Monitor	`"lcd"`	`lcd_display.py`	Full-resolution anti-aliased, system fonts

Both implement the same interface: `update_model(model)` and `set_brightness(value)`.

Backend selected by factory function in `main.py`:

```
def _create_display(backend_id, config):
    if backend_id == "lcd":
        return AtlasLCDDisplay(config)
    return AtlasLEDSimulator(config)
```

Engine State Machine (1Hz)

1. No heats → `is_armed = False`, display: “WAITING”
2. Next heat > 15 min away → shows heat info, stays WAITING
3. Next heat ≤ 15 min away → auto-activates heat (status → ‘ACTIVE’)
4. Active, before track_open → Yellow “TRACK OPENS IN” countdown
5. Active, track open → Green “ENTER NOW” countdown
6. Active, track closed, heat not ended → Red “HEAT TIME REMAINING”
7. Heat ended → status → ‘COMPLETED’, cycle repeats

Database Schema

SQLite at `data/atlas.db`

heats table:
Column	Type	Notes
—	—	—
id	INTEGER PK	Auto-increment
heat_number	TEXT	e.g. “01”
thermalling_dir	TEXT	“LEFT” or “RIGHT”
track_open	INTEGER	Unix timestamp
track_close	INTEGER	Unix timestamp
heat_end	INTEGER	Unix timestamp
status	TEXT	‘READY’, ‘ACTIVE’, ‘COMPLETED’, ‘CANCELLED’

settings table:
Key	Default	Notes
—	—	—
competition_name	“”	Event name
display_width	“128”	“128” or “64”
display_height	“64”	
led_scale	“8”	Pixel scale factor
brightness	“80”	10–100
display_backend	“simulator”	“simulator” or “lcd”
auto_boot_enabled	“0”	“0”=off, “1”=on

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4. Development Environment

Item	Detail
Machine	MacBook Air
Username	`petehep`
macOS	15.5 Sequoia
Python	3.12
Virtual env	`~/project_atlas/.venv/`
Key dependency	PySide6 6.11.1
Git	Active, many commits, `.venv/` correctly in `.gitignore`

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5. Source File Patches Applied (Session 2)

`src/core/database.py` — Cross-platform path hardening

Problem: Default `db_path="data/atlas.db"` is relative to the current working directory. Works on Mac because `run_atlas.sh` does `cd` first, but breaks on Windows if CWD is wrong.

Fix: Compute absolute path from `__file__` location:

```
def __init__(self, db_path=None):
    if db_path is None:
        _this_file    = os.path.abspath(__file__)       # …/src/core/database.py
        _core_dir     = os.path.dirname(_this_file)     # …/src/core
        _src_dir      = os.path.dirname(_core_dir)      # …/src
        _project_root = os.path.dirname(_src_dir)       # …/project_atlas
        db_path = os.path.join(_project_root, "data", "atlas.db")
    os.makedirs(os.path.dirname(db_path), exist_ok=True)
    self.db_path = db_path
```

Status: Applied on Mac ✅ | Included in Windows copy ✅

`src/gui/windows.py` — Platform-neutral exit button

Problem: Exit button in Settings tab was labelled “EXIT TO MACOS” — wrong on Windows.

Fix: Single line change, line 270:

```
# Before:
exit_btn = QPushButton("EXIT TO MACOS")
# After:
exit_btn = QPushButton("EXIT TO DESKTOP")
```

Status: Applied on Mac ✅ | Included in Windows copy ✅

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6. Mac Launcher — `run_atlas.sh`

Current content of `~/project_atlas/run_atlas.sh` (updated Session 2, all band-aids removed):

```
#!/bin/bash

# Navigate to project root (wherever this script lives)
cd "$(dirname "$0")"

# Activate virtual environment
source .venv/bin/activate

# Set PYTHONPATH so Python finds the src/ packages
export PYTHONPATH=$(pwd)/src

# Set Qt plugin path for PySide6 on macOS
export QT_QPA_PLATFORM_PLUGIN_PATH=$(python3 -c "import PySide6, os;
print(os.path.join(os.path.dirname(PySide6.__file__), 'Qt', 'plugins', 'platforms'))")
export QT_QPA_PLATFORM=cocoa

# Launch Atlas
python3 src/main.py
```

Note: Uses `python3` not `python` — macOS 15 does not have a bare `python` command.

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7. The iCloud Problem — Root Cause & Permanent Fix

What Was Happening

The project was in `~/Documents/project_atlas/` which iCloud Drive syncs automatically. Every time pip wrote `.dylib` files to `.venv`, iCloud treated it as a sync conflict and created duplicated files with broken permissions:

```
-rwx------  libqcocoa 2.dylib    ← iCloud conflict copy, wrong permissions
-rwx------  libqcocoa 3.dylib    ← another iCloud conflict copy
-rwxr-xr-x  libqcocoa.dylib      ← real file — worked until next iCloud sync
```

Qt cannot load a platform plugin with the wrong filename or permissions. This caused the `cocoa platform plugin not found` crash. Force-reinstalling PySide6 fixed it temporarily, but iCloud broke it again on the next sync.

Permanent Fix Applied

Project moved to home folder — completely outside iCloud's sync scope:

```
mv ~/Documents/project_atlas ~/project_atlas
```

iCloud only syncs `~/Desktop`, `~/Documents`, and `~/Downloads`. The home folder itself is not synced.

After the move, the venv had to be rebuilt (venv symlinks are hardcoded to creation path):

```
cd ~/project_atlas
deactivate
rm -rf .venv
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
```

If the Cocoa Error Ever Returns

First check: has the project folder somehow ended up back inside `~/Documents/`, `~/Desktop/` or `~/Downloads/`? That's the only way iCloud can touch it again.

If the project location is correct, rebuild the venv:

```
cd ~/project_atlas
deactivate && rm -rf .venv
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
./run_atlas.sh
```

Do NOT suggest force-reinstalling PySide6 as a fix — it was only ever a band-aid. The real fix was moving the project.

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8. Windows Deployment

Status

Files created, copied to Windows machine via USB stick. Windows machine setup not yet completed/verified.

Files in `windows/` Subfolder

File	Purpose
`run_atlas.bat`	Primary launcher — double-click to start Atlas
`run_atlas.ps1`	PowerShell launcher — alternative to bat file
`SETUP_WINDOWS.md`	Full setup guide — start here on a new Windows machine
`.gitkeep`	Keeps folder tracked by Git

Windows Project Location

```
C:\project_atlas\
```

Do NOT put it in Documents, Desktop or OneDrive — OneDrive causes identical problems to iCloud on Windows (will corrupt the venv).

Windows Setup Sequence (from SETUP_WINDOWS.md)

1. Install Python 3.12 from python.org — tick "Add Python to PATH"
2. Copy project to `C:\project_atlas\`
3. Delete the `.venv` folder from the copy (it's Mac-specific, useless on Windows)
4. Open CMD, navigate to `C:\project_atlas\`
5. `python -m venv .venv`
6. `.venv\Scripts\activate.bat`
7. `pip install -r requirements.txt`
8. Double-click `windows\run_atlas.bat`

Key Difference from Mac Launcher

The Windows bat/ps1 scripts set `QT_PLUGIN_PATH` (not `QT_QPA_PLATFORM_PLUGIN_PATH`) and do NOT set `QT_QPA_PLATFORM=cocoa` (that's macOS only). Windows PySide6 finds its own platform plugin automatically.

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9. Working Features (Current State)

- ✅ Real-time clock synced to system time (1Hz)
- ✅ Heat scheduling — add, edit, delete heats
- ✅ Automatic heat activation (≤15 min to track open)
- ✅ Phase-based countdown with colour coding (Yellow → Green → Red)
- ✅ LED Simulator display (128×64 and 64×64 layouts)
- ✅ LCD Monitor display (full-resolution alternative)
- ✅ Operator Console — LIVE, SCHEDULE, SETTINGS tabs
- ✅ Brightness control with live preview
- ✅ Display backend selection (restart required)
- ✅ Auto-boot toggle (defaults OFF — stay that way during development)
- ✅ EXIT TO DESKTOP button (platform-neutral label)
- ✅ RESTART ATLAS button
- ✅ Abort/Cancel active heat
- ✅ Settings persist across restarts (SQLite)
- ✅ Cross-platform path handling for database

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10. Known Issues & Open Threads

Open

- [ ] **Windows machine setup not yet verified** — SETUP_WINDOWS.md has the steps, needs to be walked through on the actual Windows PC
- [ ] **`atlas.sh`** at project root — may be an outdated duplicate of `run_atlas.sh`, worth checking and removing if redundant

- [ ] `core/timers.py` **and** `core/config.py` — legacy files, not used by any active code. Could be cleaned up but harmless to leave
- [ ] `atlas_config.json` — legacy JSON config, not used by `main.py`. Harmless but vestigial

Future / Roadmap

- [ ] Physical LED hardware integration (P5/P10 panels, HUB75 protocol)
- [ ] macOS LaunchAgent for appliance-mode auto-start
- [ ] Additional display backends (Artnet DMX, Serial/UART)
- [ ] Pre-loaded competition schedules

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11. Documentation Set

All documents follow the `ATLAS-XXX-001` naming convention. The authoritative set as of July 26, 2026:

Document	File	Notes
README	`README_ATLAS.pdf`	Project overview
Architecture	`ATLAS-ARC-001_Current_Architecture_and_System_Status.pdf`	System architecture
Actual System Review	`ATLAS-ASR-001_Actual_System_Review_v2.pdf`	What's actually built
Deployment Guide	`ATLAS-DEP-001_Deployment_Guide_v2.pdf`	Use v2, not v1
Document Register	`ATLAS-DOC-001_Document_Status_Register.pdf`	Doc index
Progress Log	`ATLAS-LOG-003_Progress_Status_July2026.pdf`	Latest progress
Interface Control	`ATLAS-ICD-001_Interface_Control_Document.pdf`	Signal/interface spec
Mac PySide6 Fix	`MAC_PYSIDE6_FIX.md/.pdf/.docx`	iCloud root cause + fix
Windows Setup	`windows/SETUP_WINDOWS.md/.pdf/.docx`	Windows install guide
This document	`ATLAS_PROJECT_HANDOVER_Updated.md`	You are here

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12. Quick Reference

Launch on Mac

```
cd ~/project_atlas && ./run_atlas.sh
```

Launch on Windows

Double-click `C:\project_atlas\windows\run_atlas.bat`

If Atlas Won’t Start on Mac

```
cd ~/project_atlas
deactivate && rm -rf .venv
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
./run_atlas.sh
```

Key File Locations

What	Where
Project root (Mac)	`~/project_atlas/`
Project root (Windows)	`C:\project_atlas\`
Mac launcher	`~/project_atlas/run_atlas.sh`
Windows launcher	`C:\project_atlas\windows\run_atlas.bat`
Entry point	`src/main.py`
Engine	`src/core/engine.py`
Database manager	`src/core/database.py`
Operator Console	`src/gui/windows.py`
LED Simulator	`src/gui/led_simulator.py`
LCD Display	`src/gui/lcd_display.py`
Database file	`data/atlas.db`
Virtual env (Mac)	`.venv/` — do NOT commit, do NOT let it end up in iCloud/OneDrive

Settings That Need a Restart

- Display backend (LED Simulator ↔ LCD Monitor)
- Display size (128×64 ↔ 64×64)
- Auto-boot toggle

Settings That Apply Immediately

- Brightness slider (live via signal)

End of handover. Future AI: read Peter’s actual files before changing anything.
Confirm what’s running on his machine vs what’s in any workspace.
If you broke something, say so and fix it — don’t dress it up.